# TopoWorld-Eval: Grounded, Coherent, and Revisable 3D Worlds from Multi-View Architectural Evidence

Jincheng Li Yuxuan Li

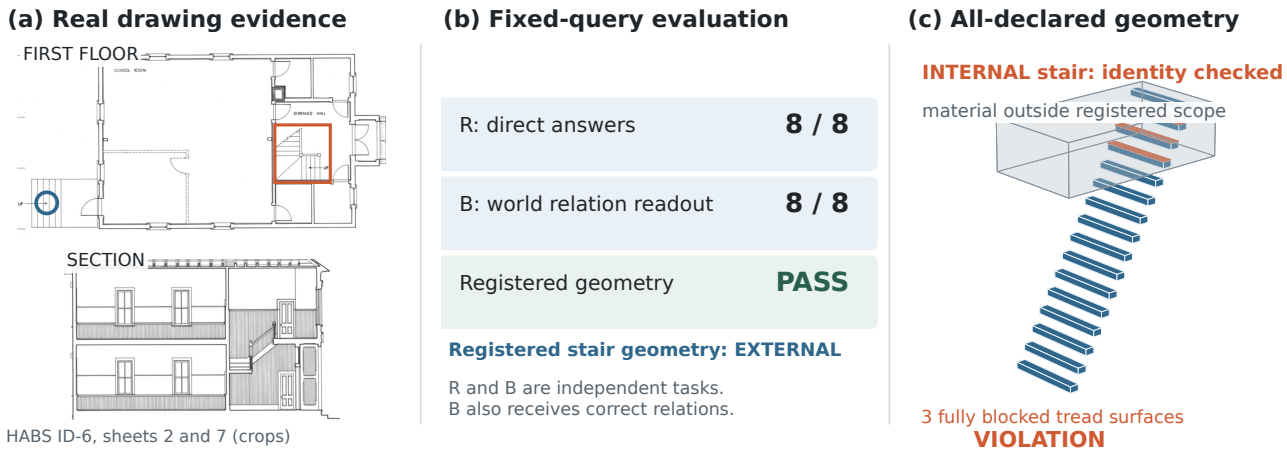



Figure 1. **One source, different spatial tests.** Real drawings of Idaho City Schoolhouse define the input. Route B receives correct determinate relations in addition to the images. Its preserved GPT-5.6 Luna world realizes all eight fixed queries and passes registered geometry, yet its internal staircase intersects the declared slab. The external stair geometry is registered; the internal stair's identity is queried, but its material is outside that scope. Source grounding, fixed readouts, and material coherence are separate tests.

## Abstract

*Multimodal models can answer spatial questions without producing a coherent executable world. We introduce* TopoWorld-Eval, *an evaluation of whether models construct and maintain grounded, geometrically consistent, and revisable worlds from multiple architectural drawings. It contains 14 buildings, 42 drawing pages, 95 determinate relations, and one unresolved identity relation. We compare direct answering, independent world construction, and construction assisted by correct relations. Across eight GPT product configurations, the 12-building point-anchored condition yields 98.17% answer accuracy and 88.50% world-readout accuracy. Among jointly contract-valid opportunities, answering alone is correct in 34 cases and construction alone in five. Correct relations rescue 29 construction errors but introduce 11. Qwen largely aligns answering and construction in N2, while complex stairs expose interface failures. We also measure grounding, material consistency, and evidence withdrawal. Spatial answering, executable construction, and world maintenance require separate measurements.*

## 1. Introduction

Our question is: *Can multimodal models maintain a grounded, coherent, and revisable 3D world across multiple views and tasks?* A spatial assistant should refer to the same building when it answers, constructs, and revises. A stair seen in a plan must match the intended stair in a section. Its generated endpoints must meet the right floor surfaces. A window must keep its host across views. When a new section resolves a height ambiguity, the model should update that fact while preserving other supported relations.

These requirements are related, but they are different tests. A correct sentence about a staircase does not establish an executable connection. A geometrically consistent program can describe the wrong building. Even a world that satisfies every fixed query may contain a conflict in additional generated material (Fig. 1).

TopoWorld-Eval connects these measurements on the same architectural evidence. For fixed targets, route R answers relations directly; route A constructs an executable world independently; route B constructs from the same evidence plus correct relation values. A deterministic reader derives answers from the generated geometry and instance bindings. It does not read route R. We separately check the registered target fragment and everything the model additionally declares.

Our bounded interface supports rooms, hosted openings, walls, slabs, straight stair runs, and assemblies with landings. It permits multiple geometric completions and separates invalid syntax, wrong bindings, incompatible interfaces, material obstructions, and uncertainty.

The experiments cover eight GPT product configurations at recorded `xhigh` effort and a fixed Qwen snapshot. The main 12-building N2 condition gives anonymous target points; the two-building CS1 supplement also specifies the target unit, such as a flight or assembly. These are given-target tasks, not full-page object discovery. The two providers use different interfaces, so the comparison concerns declared workflows rather than isolated neural weights.

The experiments measure when these observable outputs align across input conditions and provider workflows. Correct relations rescue some construction errors and introduce others. Explicit knowledge updates can leave retained spatial relations changed. R and A use independent contexts, so their difference concerns task performance rather than direct evidence of a shared internal state.

**Contributions.** (1) We introduce a benchmark that links source-grounded relation answering, executable construction, and evidence revision through shared multi-view building evidence. (2) We define separate measures of relation realization, registered and all-declared geometry, material obstruction, image grounding, and evidence retention through a typed world interface and deterministic readouts. (3) We provide an empirical analysis of preserved first outputs across model configurations, input conditions, and two provider workflows, including successful alignment, assisted rescues and regressions, contract failures, and measurement limitations.

## 2. Related Work

**Spatial questions and multi-view understanding.** OpenEQA evaluates questions about environments from remembered observations or active exploration [7]. VSI-Bench studies video-based spatial recall and reasoning, including cognitive maps [12]. MMSI-Bench evaluates multi-image spatial intelligence, while MindCube examines mental modeling from limited views [10, 13]. SpatialScore expands the range of spatial tasks and modalities [11]. These benchmarks cover varied spatial tasks. TopoWorld-Eval links answers about fixed targets to independently constructed, executable artifacts from the same architectural evidence. It measures whether the artifacts realize those relations, retain the intended image bindings, remain materially coherent, and respond to evidence changes. These output properties are not determined by answer accuracy alone.

**Structured reconstruction and scene generation.** Explicit scene representations are well established. SceneScript predicts structured commands that a geometry interpreter converts into a scene [1]. SpatialLM generates architectural elements and object boxes from point clouds, and CAD-Recode generates executable Python CAD programs [8, 9]. These systems recover inspectable scene or object structure through explicit geometric representations. Holodeck uses language-derived spatial constraints and layout optimization; PhyScene incorporates collision, layout, and reachability guidance [14, 15]. Holodeck 2.0 adds vision-language-guided generation and interactive editing [2]. TopoWorld-Eval uses inspectable geometry as an evaluation interface for architectural drawings. Different coordinate completions are allowed when they realize the same evidence-supported target obligations. The measurement links independent task outputs and checks the generated materials, interfaces, and observation bindings; it does not require one reference surface reconstruction.

**Geometric constraints, feedback, and revision.** Geometrically-Constrained Agent already identifies a semantic-to-geometric gap and formalizes reference frames and objectives before constrained geometric computation [3]. 3DThinker uses geometric latent representations for reasoning from limited views [4]. These methods motivate a distinction between geometric reasoning mechanisms and the outputs used to evaluate them. TopoWorld-Eval measures independently produced answers and world artifacts, including whether supplied correct relations are realized without an invalid interface or material conflict. Self-Refine establishes feedback-based iterative improvement [6]; our main R/A/B routes preserve first outputs without checker feedback. Source withdrawal poses a separate question from scene editing: which conclusions remain supported after the evidence changes? Truth maintenance provides the established foundation of recorded justifications and invalidated assumptions [5]. Our revision test makes these effects observable by separating measured knowledge from hypothetical geometry and comparing the updated world with a fresh construction from the same effective evidence.

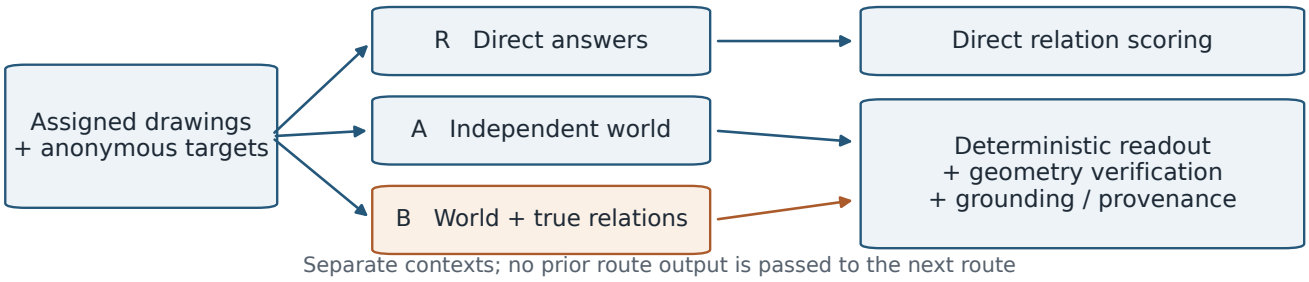



Figure 2. Independent routes use the same assigned drawings and targets. Only B receives reference relation values; it receives no reference coordinates. World readout and geometry verification occur outside model generation. A separate sequential subset adds and withdraws one source, then compares the revised world with a fresh-context reconstruction.

## 3. Benchmark Design

### 3.1. Sources, targets, and references

The dataset contains 14 buildings and 42 whole drawing pages from public measured-building records, including plans, elevations, and sections. Twelve buildings provide 75 determinate relations and one unresolved identity relation. Lovell House and Gamble House add 20 determinate relations involving more complex stair fragments. Source pages, public view permutations, target locations, and reference rationales are retained with hashes. We do not treat pages or repeated tasks as new buildings.

The 95 determinate relations cover above/below, same-floor membership, separation, door and window hosts, connectivity, and cross-view identity. References were assembled by AI-assisted source inspection, cross-checked in a separate AI review, and subsequently reviewed by the author across all 96 obligations. They specify supported relations and uncertainty, not a complete measured 3D model.

The unresolved example concerns a staircase at the same location in drawings from different periods. Position alone does not establish that the physical object was never replaced. Its reference remains UNKNOWN. Different image IDs, repeated type numbers, or similar projections do not themselves decide physical identity.

### 3.2. Input conditions

**N2: point-anchored understanding.** For 12 buildings, anonymous targets receive normalized image points and class hints. Models must infer extent, instance correspondence, floor surface, and host from the pixels. Printed image labels are allowed. Reference boxes, private bindings, and external descriptions are excluded.

**CS1: point and target-unit anchors.** Two new buildings also specify whether a target is one straight flight or an assembly between adjacent primary floor plans. The unit information does not supply the correct cross-view mapping. Because it changes the input, CS1 is reported separately from N2.

**Historical v1.** An earlier pass on the same 12 N2 buildings used richer semantic locators that could cue room or host information. It remains a development input control, not an additional independent building set. Its results are reported in the supplement.

### 3.3. Three routes on the same evidence

**R—direct answering** returns TRUE, FALSE, or UNKNOWN for fixed queries with source references. **A—independent construction** returns world JSON under the public contract; the evaluator reads the queries from this artifact. **B—relation-assisted construction** additionally receives correct determinate relation values, but no reference geometry. All three are independent contexts. A never reads R, and B never reads A. Their comparisons test task performance and the effect of relation assistance, not a common hidden state being translated into coordinates.

### 3.4. Scope and ambiguity

Targets define a fragment rather than the whole building. Multiple geometries can realize the evidence, so unmeasured dimensions may be marked as hypothetical completions. Missing geometry does not become a pass: required coverage and query readouts are still checked. Data are public drawings rather than RGB walkthroughs; pretraining exposure is unknown. Building identities are distinct, but layout-family independence is not established.

# 4. Executable Worlds and Evaluation

## 4.1. A bounded executable interface

We summarize the implemented world as

$$W = (\mathcal{O}, \mathcal{G}, \alpha, \mathcal{R}, \mathcal{P}). \tag{1}$$

| Field | Executable meaning |
|---|---|
| $\mathcal{O}$ | Physical instances, parts, assemblies, interfaces |
| $\mathcal{G}$ | Occupied or free regions in a shared coordinate frame |
| $\alpha$ | Observation-to-instance bindings, including unresolved bindings |
| $\mathcal{R}$ | Typed declarations: hosts, floors, identity, connections |
| $\mathcal{P}$ | Sources, active status, and explicit support edges |

Table 1. World fields connect physical identity, executable geometry, and evidence support in one inspectable artifact.

Boxes and box unions represent material, free space, openings, or interfaces. Straight stair runs expand into tread geometry; finite assemblies declare runs and landings. The assembly material must equal its parts' material union. Transforms are restricted to supported signed-axis permutations and translations. Unknown geometry is explicit.

Rooms represent spatial regions, not solid obstacles. Walls, slabs, and treads occupy material. Hosted openings remove material rather than merely overlaying a rectangle. A stair's connection labels do not prove contact: its endpoint interfaces, landing top surfaces, host spaces, and run chain must satisfy the corresponding geometric tests. Observation identity is read from $\alpha$ and compared with the source-grounded reference.

## 4.2. Separate measurements

**Representation validity** distinguishes delivery, character/entity limits, JSON, Schema, and compilation. The frozen output budget is 42,000 characters and 60 entities; first outputs are preserved without repair. Unsupported or unknown geometry is not counted as a definite physical error.

**Relation realization** compares direct R answers or deterministic world readouts with the fixed reference queries. For each route,

$$\mathrm{Acc} = \frac{\#\,\text{correct determinate readouts}}{\#\,\text{allocated determinate opportunities}}. \tag{2}$$

Missing and contract-invalid outputs remain in this denominator. Reference-unknown identity is evaluated separately for appropriate uncertainty. A definite answer there is unsupported certainty, not necessarily a provably false physical claim.

**Geometry coherence** has two scopes. Registered geometry checks geometric target roles and their dependency closure. All-declared geometry also checks additional generated structures. Material-obstruction tests identify declared material blocking supported tread/interface regions; they do not certify full body clearance. For scope $\Gamma$, joint success is

$$\begin{aligned} J_\Gamma = \mathbf{1}[&V \wedge C \\ &\wedge (\forall q \in K : \widehat{y}_q = y_q) \\ &\wedge \mathrm{Geom}_\Gamma(W) = \mathrm{PASS}], \end{aligned} \tag{3}$$

where $V$ is contract validity, $C$ required coverage, and $K$ the determinate queries. We also report each component, because one scalar cannot distinguish missing geometry from an obstruction.

**Grounding and revision** are separate. Given-point localization uses observation boxes and exact source IDs at IoU$\geq 0.25$ against approximate reference boxes. Revision checks explicit measured knowledge, active sources and dependencies, and retained spatial readouts under current evidence.

## 4.3. Instrument qualification

N2 and CS1 prospectively use evaluator 2.2.1. Qualification includes 83 unit tests, 29 independent re-encoding/counterexample checks, and a 200-case blind-hole diagnostic. These are software checks, not building samples. Historical v1 was generated under 2.2 and uniformly reassessed; its score change is not model improvement. Original outputs and scores remain intact. A later slab-encoding limitation is reported with the results rather than selectively changing one model's score.

| Input | R: answers | A: worlds | B: assisted worlds |
|---|---|---|---|
| N2 | 589/600 (98.17%) | 531/600 (88.50%) | 537/600 (89.50%) |
| CS1 | 146/160 (91.25%) | 115/160 (71.88%) | 125/160 (78.13%) |

Table 2. **GPT relation realization on allocated opportunities.** Eight product configurations are evaluated at `xhigh`. All delivery and contract failures remain in the denominators. N2 gives target points; CS1 additionally specifies target units and uses two different buildings. The rows are not matched-difficulty tests.

| Input | Pair | $n$ | Both | First only | Second only | Neither |
|---|---|---|---|---|---|---|
| N2 | R $\rightarrow$ A | 570 | 526 | 34 | 5 | 5 |
| N2 | A $\rightarrow$ B | 533 | 487 | 11 | 29 | 6 |
| CS1 | R $\rightarrow$ A | 140 | 113 | 15 | 2 | 10 |
| CS1 | A $\rightarrow$ B | 140 | 110 | 5 | 15 | 10 |

Table 3. **Both directions of paired change.** Only pairs with both output contracts valid enter this table. First/second only refers to the route order in the Pair column. Valid JSON can still contain unevaluable geometry. The full-denominator results remain in Table 2.

# 5. Main Experimental Results

## 5.1. Configurations and first-output accounting

The eight GPT configurations are GPT-6 Astra; GPT-6.1 Sol and Terra; GPT-6 Sol and Luna; and GPT-5.6 Sol, Terra, and Luna. These are product-visible aliases within one provider family; an immutable backend weight revision is not exposed. Codex permits reading assigned files and images, cropping those pixels, and saving the first response. The protocol excludes hidden references, other model outputs, solvers, checker feedback, search, and delegated solving.

The preserved campaign has 688 task stages: 288 historical v1 R/A/B, 288 N2, 48 CS1, and 64 revision stages. Of the stages, 651 satisfy the output contract. Standard raw paths contain 684 files, one empty. Missing, malformed, overlimit, and wrong-location outputs are retained. Observed access departures and technical recovery diagnostics remain separate from primary scores (supplement).

## 5.2. Answering and construction differ

In N2, R achieves 589/600 and A 531/600. On the 570 opportunities where both contracts are valid, R alone is correct in 34 cases and A alone in five. The paired difference $A - R$ is $-5.09$ percentage points. Thus the difference is not explained solely by JSON or delivery failure. A readout may still fail through geometry, an invalid interface, or uncertainty; it is not automatically a hidden visual error.

An exploratory building-cluster bootstrap uses 20,000 resamples, retaining all configurations and queries of each sampled N2 building. The 95% interval for $A - R$ is $[-9.29, -1.55]$ points. It describes variation over these 12 buildings, not unseen model or deduplicated layout families.

## 5.3. Correct relations help and hurt

Within 533 jointly valid A/B opportunities, B rescues 29 errors and introduces 11: a net increase of 18, or 3.38 points, with an exploratory cluster interval $[0.19, 6.42]$. On all 600 opportunities, B improves by only six correct readouts. Unequal contract-failure sets affect the allocated comparison. The common-valid comparison isolates content changes within a shared subset.

CS1 has 15 R-only and two A-only successes in 140 common valid opportunities. B rescues 15 and damages five. With only two buildings, we report these counts without a stable population confidence claim.

## 5.4. Reading the high answer scores

N2 has 61 TRUE and 14 FALSE references, so an always-TRUE answer reaches 81.33%. CS1 has ten of each, giving 50%. These controls describe answer-label imbalance, not world construction. The supplement gives per-configuration and per-building outcomes; eight configurations are not eight independent model families.

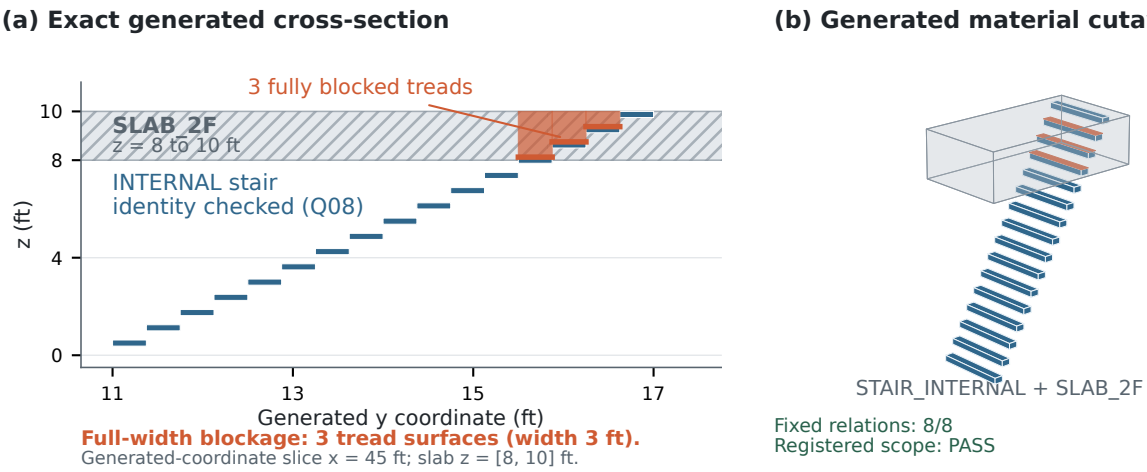



Figure 3. **A real conflict outside registered geometric scope.** The preserved N2/Idaho City/B world (GPT-5.6 Luna) realizes all eight fixed relations and passes registered checks. Its additional `STAIR_INTERNAL` rises from $z = 0$ to 10 ft while `SLAB_2F` occupies $z \in [8, 10]$ ft without an opening. The frozen proof identifies three full-width blocked tread surfaces. The registered connection concerns a different, external stair. This is an all-declared material conflict, not a failure of that registered external-stair relation. The internal stair's identity is queried, but its material is outside the registered geometric scope.

## 6. Geometric Coherence and Failure Analysis

| Input/route | $n$ | All rel. | Reg. | Decl. | Block. |
|---|---|---|---|---|---|
| N2/A | 96 | 69 | 50 | 16 | 1 |
| N2/B | 96 | 78 | 49 | 13 | 2 |
| CS1/A | 16 | 3 | 4 | 1 | 1 |
| CS1/B | 16 | 5 | 4 | 2 | 1 |

Table 4. World-level counts. $n$: allocated worlds; All rel.: all determinate queries correct; Reg./Decl.: registered/all-declared geometry PASS; Block.: definite material obstruction. These columns measure different events.

### 6.1. Correct relations do not exhaust a world

All three N2 obstruction worlds satisfy every fixed query. Two are Drayton Hall A/B; the third is Fig. 3. These cases distinguish fixed-query coverage from consistency of the submitted world. A model can realize the queried relations and create a conflict in additional material. Registered and all-declared scopes make the distinction visible.

N2 joint registered success (Eq. 3) is 42/96 for A and 49/96 for B; joint all-declared success is 14/96 and 13/96. For CS1, the corresponding counts are 1/16 versus 3/16, and 1/16 versus 1/16. No PASS certifies the complete real building.

### 6.2. Local level changes and assemblies

Lovell Q04 involves a two-riser UP transition rather than an ordinary story change. GPT R succeeds 0/8, A 2/8, and B 5/8. Gamble's ordinary floor distinction in Q01 instead yields 8/8, 8/8, and 7/8. The contrast locates a difficulty in this local case; it does not show a general failure to identify floors.

Assemblies have obligations beyond labels: parent and part materials must agree, landings must support endpoints, and space entities cannot silently replace landing interfaces. These checks distinguish an unrealized connection from a plausible description.

### 6.3. Instrument errors remain distinct

A later slab-opening diagnostic found a 2.2.1 limitation. In a predeclared ten-world cohort, eight re-encodings preserved final material but changed a detached-opening violation label; two changed material and do not establish an error. The eight became unevaluable, not PASS. Fixed readouts and registered verdicts were unchanged, and historical A/B declared-PASS counts did not increase. We preserve the primary version and disclose this diagnostic rather than repairing selected outputs.

## 7. Cross-Family Results and Grounding

| Input | R | A | B |
|---|---|---|---|
| N2 | 69/75 | 62/75 | 69/75 |
| CS1 | 19/20 | 16/20 | 16/20 |

Table 5. Qwen `qwen3.8-max-0902` on allocated determinate opportunities. Of 42 unique first submissions, 39 returned complete responses and 38 satisfy the output contract. Transport and Schema failures remain in these denominators.

### 7.1. A model–interface comparison

Qwen receives the same 14 buildings and R/A/B core tasks through a tool-free multi-image API. Thinking is enabled with a 32,768-token budget, completion is capped at 49,152 tokens, and temperature is 0.6. Its R wrapper additionally contains the generic world Schema, whereas GPT-R does not. This Schema contains no building answers. GPT can crop assigned local pixels; Qwen uses submitted image inputs. The settings and tools are not compute-matched.

Two initial non-streaming requests disconnected without final output. They remain failures, not retried replacements. The remaining 40 tasks each received one first streaming submission; one also disconnected. The comparison includes the actual delivery conditions, not just successful replies.

### 7.2. N2 mostly aligns answers and worlds

In N2, R is correct on 69/69 contract-valid opportunities, A on 62/63, and B on 69/70. Of 63 common-valid R/A opportunities, 62 are jointly correct and one is R-only. A/B share 58 valid opportunities: 57 jointly correct and one jointly wrong, with no rescue or regression. The seven-readout increase on the full denominator from A to B comes from different failure sets, not paired content improvement.

The single R/A mismatch is a school stair connection whose generated endpoint interface is a corridor, not a landing. This is a contract-level instantiation failure, not established evidence of wrong physical identity. The strong N2 alignment shows that the observed answer–construction difference varies across model–interface conditions.

### 7.3. CS1 exposes stair implementation failures

All six CS1 outputs are contract-valid. R/A/B yield 19/20, 16/20, and 16/20. All eight determinate identity relations per route are correct. Three A readouts fail despite correct R answers: a Lovell landing declares `floor_z`=5.29 ft while its material top is 5.79 ft; a Gamble assembly's material differs from the union of its parts; another endpoint directly uses a corridor instead of a landing.

B repairs the two-riser floor relation that R and A both missed, but damages a previously correct endpoint relation. A/B thus have one rescue and one regression despite equal totals. No definite material obstruction is detected in the 25 valid Qwen construction worlds.

### 7.4. Grounding and insufficient evidence

Grounding links world instances back to the assigned drawings. For GPT N2, given-point box IoU$\geq$ 0.25 holds in 446/656 A records and 440/656 B records (mean IoU 0.491/0.489); 16 abstract-region references per route are not evaluable. CS1 yields 49/88 and 40/88. This is not full-page detection or complete 3D grounding.

Strict source-ID matching makes some Qwen boxes score zero: 56 existing boxes use aliases such as `S_V03` for `V03`. Under a lexical-alias diagnostic assumption, 55 reach the IoU threshold. This does not replace primary scores. Other failures lack target boxes or have low overlap; low IoU alone cannot establish that an object was not seen.

On unresolved identity, GPT N2 preserves UNKNOWN only 2/24 times, versus 18/24 in v1. Qwen R/A assert identity and B fails to return: two unsupported certainties plus a service failure, not three proven identity errors.

## 8. Evidence Revision and World Maintenance

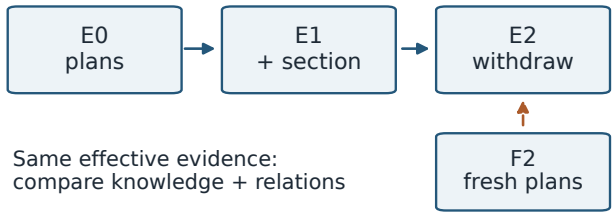



Figure 4. Evidence arrival and withdrawal differ from physical building change. E2 and fresh F2 use the same effective sources; retained knowledge and relation readouts are compared.

The revision subset uses two buildings, eight configurations, and four stages, totaling 64 stages. E0 receives plans; E1 adds a section supporting an exact height; E2 withdraws it in the same context. F2 constructs afresh from E2's effective evidence. The three sequential stages are not independent repetitions.

| Stage | Valid | Knowledge | Relations |
|---|---|---|---|
| E0 | 15/16 | 15/15 | 68/80 |
| E1 | 10/16 | 10/10 | 49/80 |
| E2 | 13/16 | 13/13 | 58/80 |
| F2 | 14/16 | 14/14 | 65/80 |

Table 6. Revision accounting. Valid: contract-valid outputs; Knowledge: correct explicit status among valid outputs; Relations: retained correct readouts over all allocated opportunities.

All 52 valid stages have correct explicit height-knowledge states, active sources, and support edges. All ten valid E1 worlds also implement the supplied height. Withdrawal may return a measured fact to unknown while leaving hypothetical completion coordinates: those are different claims.

E2 and F2 have 11 jointly valid building–configuration pairs. Their knowledge states agree 11/11, but retained readouts agree 51/55. E2 alone is wrong in three cases and F2 alone in one. Explicit source-state accuracy and spatial retention therefore need separate measurements. These differences do not by themselves identify stale neural memory as their cause.

## 9. Discussion and Limitations

**One question, several observable capabilities.** High R accuracy, a valid program, and a coherent fragment are different achievements. Correct relations can improve construction without ensuring valid interfaces or harmless additional material. Qwen's N2 success also shows that the difference is not uniformly large. The benchmark measures when these outputs align and diverge.

**Scope of the evidence.** The dataset is small and target-assisted. CS1 adds unit information; N2 reuses v1 buildings; revision is not repeated under N2/CS1. Public pretraining exposure and layout-family dependence are unresolved. References describe selected drawing-supported relations, not complete site measurements. Different provider tools and reasoning budgets cannot isolate neural weights.

**Scope of geometric checks.** The compiler supports finite box/stair fragments, not general BIM, arbitrary meshes, curved stairs, structural safety, or full human clearance. Undeclared obstacles remain outside its scope. The slab equivalence issue is an instrument limitation; future corrections should be versioned and applied uniformly.

**Scope of revision.** Removing a source ID does not prove that information has disappeared from candidate domains, bindings, initialization, or hidden state. Identity and scope corrections need further task variants. Future comparisons should distinguish them and include ordinary reconstruction and source-tracking controls where applicable.

## 10. Conclusion

TOPOWORLD-EVAL evaluates whether answers, executable geometry, grounding, and evidence updates remain compatible with one observed world. The preserved experiments show condition-dependent task differences, both benefits and regressions from correct relations, material conflicts outside fixed queries, and a separation between explicit knowledge updates and retained spatial conclusions. Joint measurement gives a more specific account of spatial understanding than answer accuracy or geometric validity alone.

## LLM Usage Statement

OpenAI Codex with GPT models was used extensively for research development: task and representation design, code generation and debugging, source-drawing inspection, AI reference cross-checks, experiment orchestration, analysis, figure-generation code, related-work searches, and manuscript drafting and language revision. The author directed the research and reviewed all 96 reference obligations. The eight GPT product configurations and Qwen `qwen3.8-max-0902` are also evaluated systems, as specified in the experimental protocol. Their first outputs and failures were preserved. Paper figures were rendered from source drawings, recorded data, and preserved geometric outputs; no generative image model fabricated experimental evidence. The author remains responsible for the claims, references, and submitted artifacts.

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